Nonlinear & Dynamic Finite Element Analysis



Finite Element Analysis

- Nonlinear Finite Element Analysis
- Dynamic Finite Element Analysis
- Explicit Time Integration
- Implicit Time Integration



Finite Element Analysis

- **Nonlinear Finite Element Analysis**
- Dynamic Finite Element Analysis
- Explicit Time Integration
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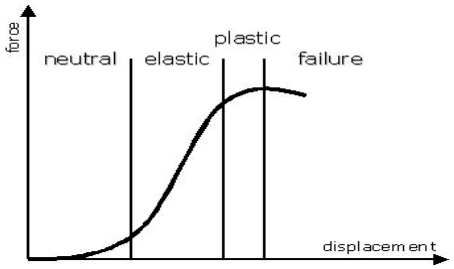
Linear analyses

$$\mathbf{KU} = \mathbf{F}$$

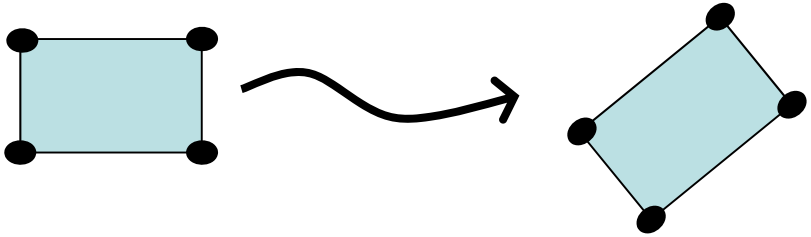
- **U** is a linear function of **F** ($\alpha\mathbf{F} \Rightarrow \alpha\mathbf{U}$)
- Small displacements in evaluation of:
  - **K**
  - **F**
- All integrations performed over original volume of the finite elements.
- Strain-displacements matrix **B** assumed to be independent of element displacements.
- Linear elastic material is implied.
- Unchanged boundary constraints $\Rightarrow$ constant BC

Nonlinear analyses

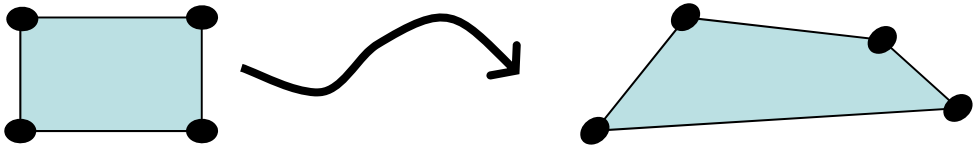
- Material nonlinearity

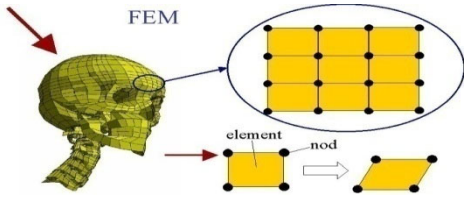


- Large displacements & rotations

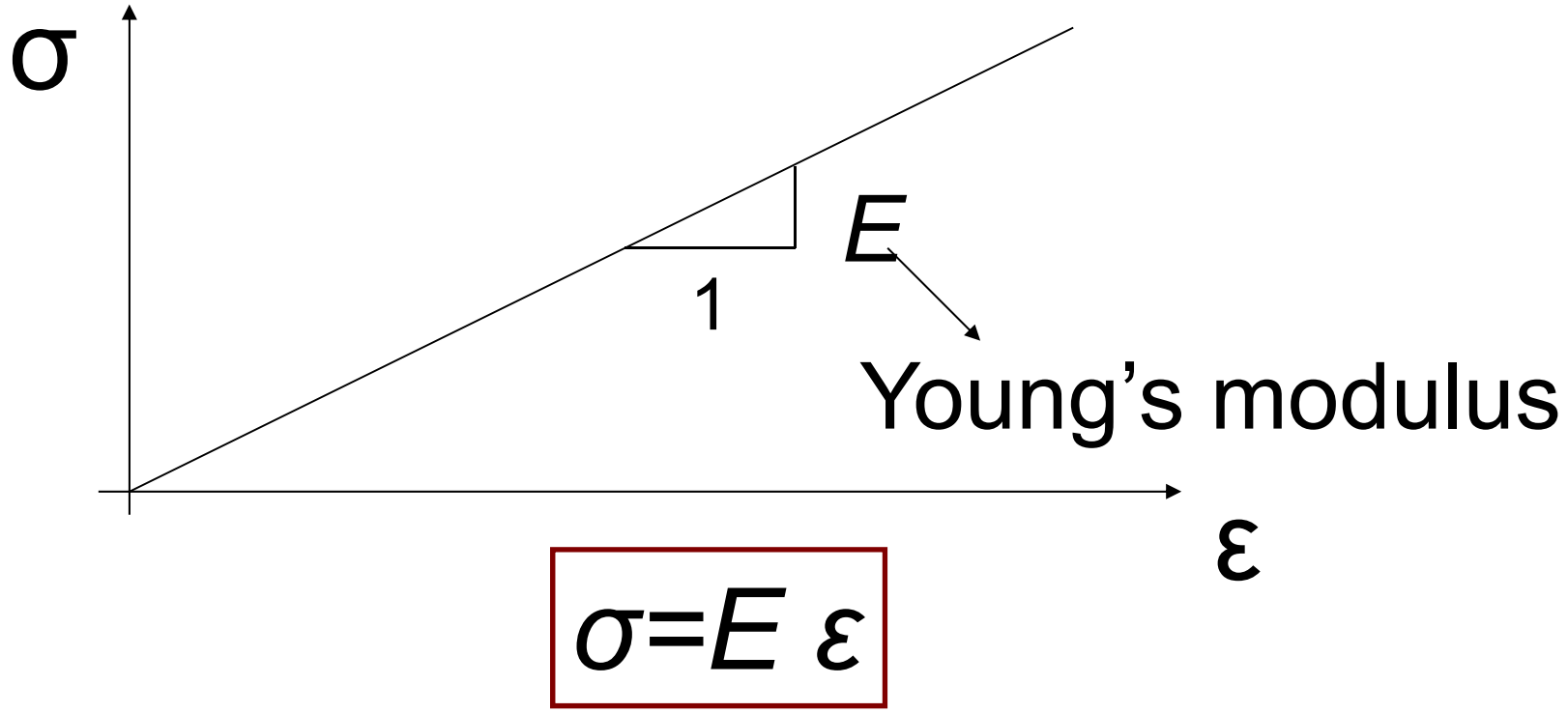


- Large deformations (large strains)





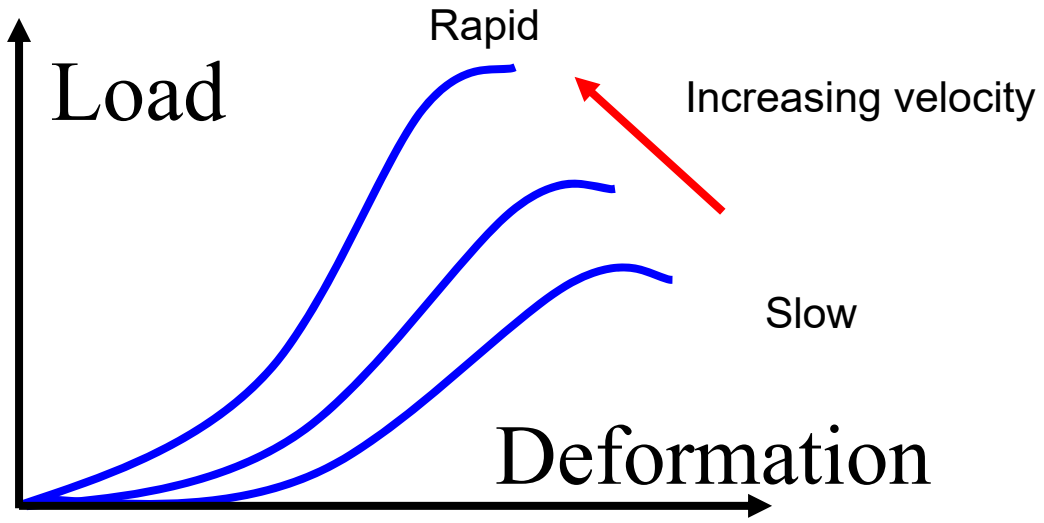
Hooke's law



$$\sigma = E\,\varepsilon$$

Biological tissues

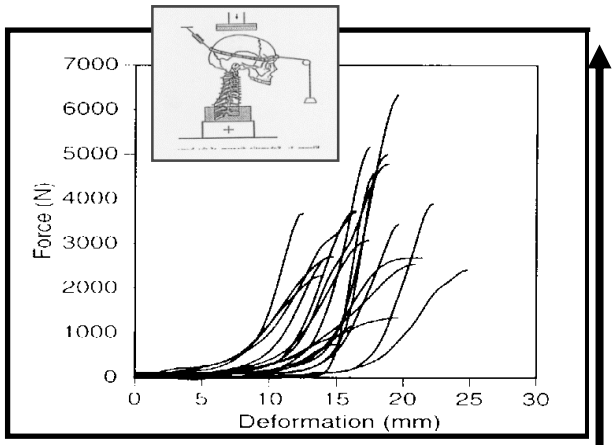
- Non-linear
- Anisotropic
- Inhomogeneous
- Viscoelastic

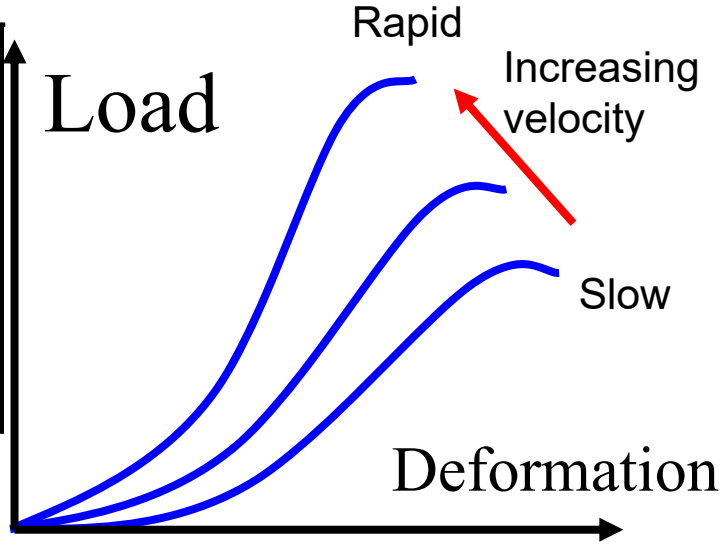


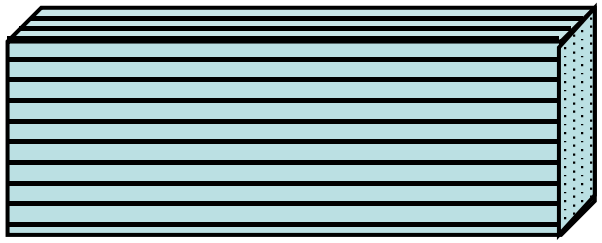


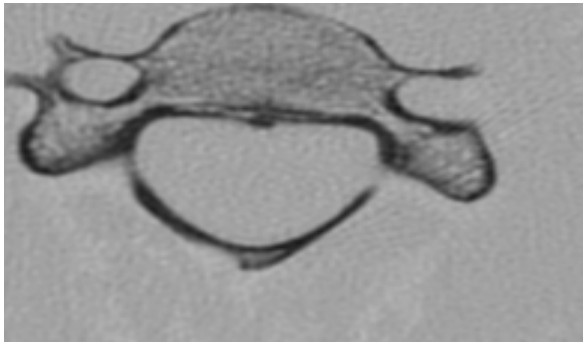
Biological tissues

- Non-linear
- Anisotropic
- Inhomogeneous
- Viscoelastic









Soft Tissues of the Human Body

Skin

Tendons

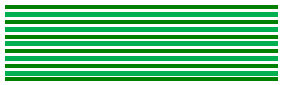
Ligaments

Cartilage

Vascular tissue

Intervertebral Discs

Composites



Reinforcement:

Collagen

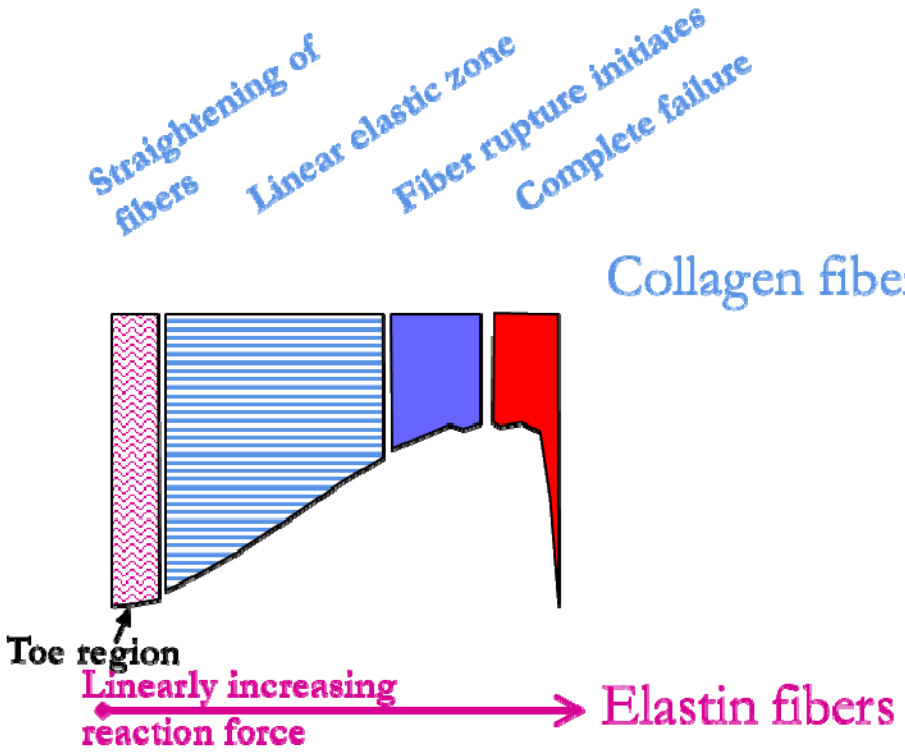
Elastin

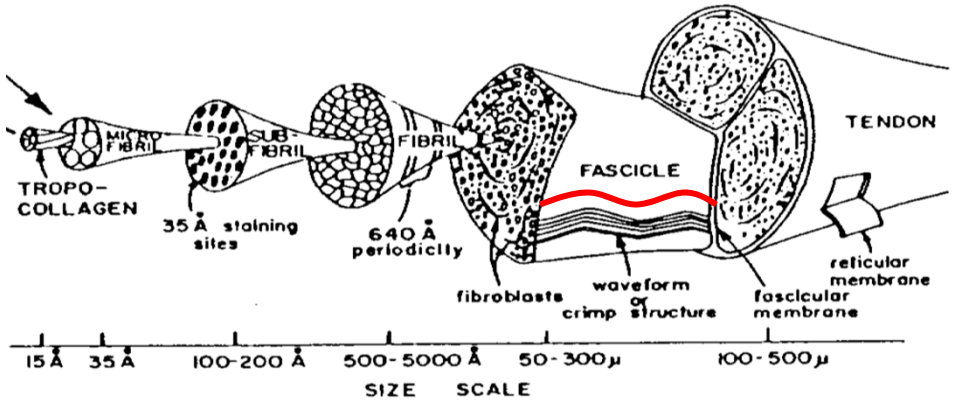
Ground Substance:

Proteoglycans

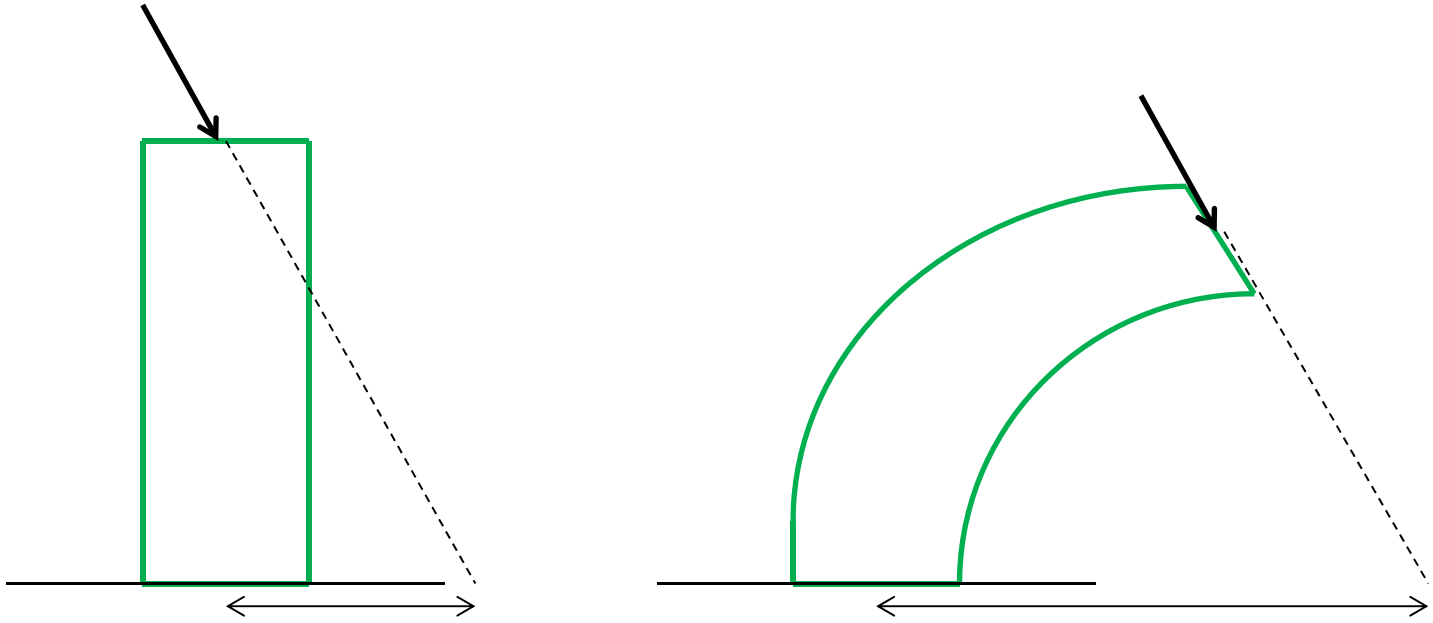
Tissue Fluid

Soft tissue nonlinear elasticity

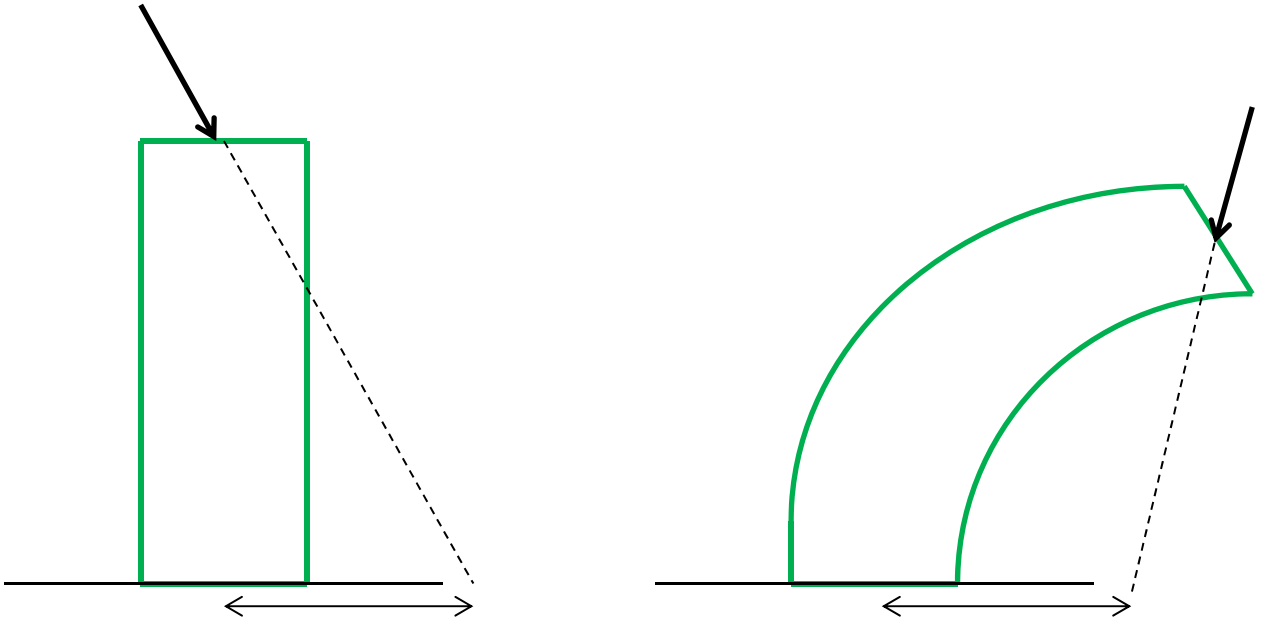




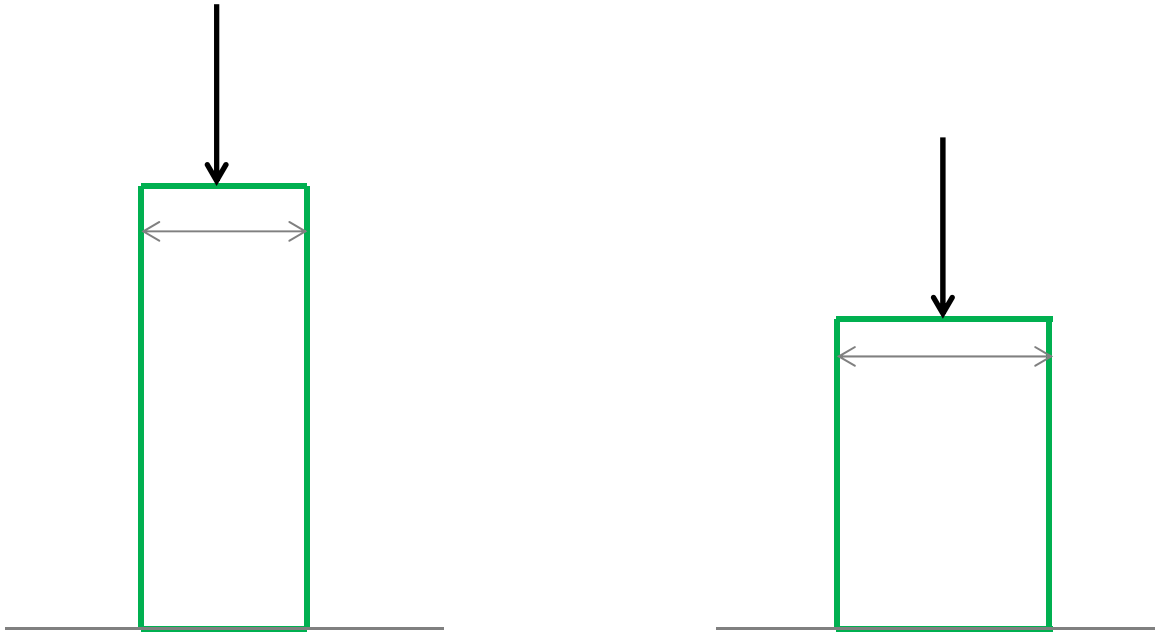
Assuming small displacements in evaluation of the load



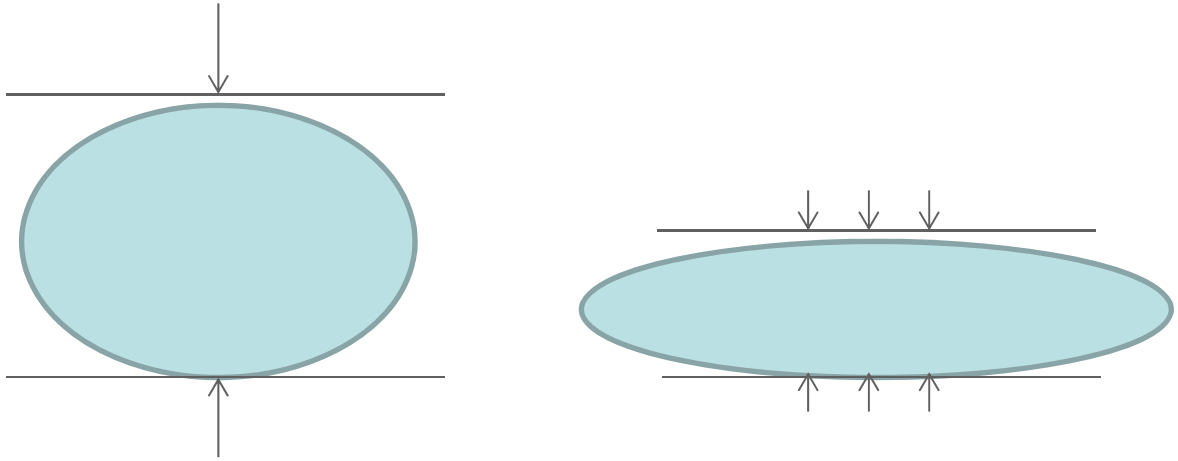
Accounting for large displacements/rotations



Assuming no change in surface and volumes



Changed loading surface due to deformation



Changed loading surface due to displacement



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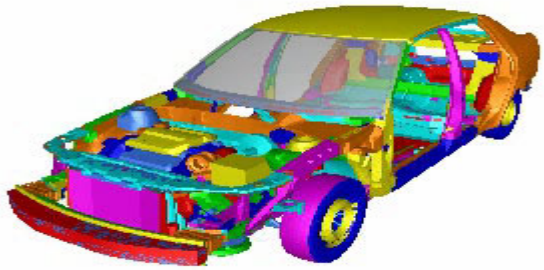


When is a problem dynamic?

- If mass inertia effects the responses.
- Frequency of applied loading > $\omega/5$

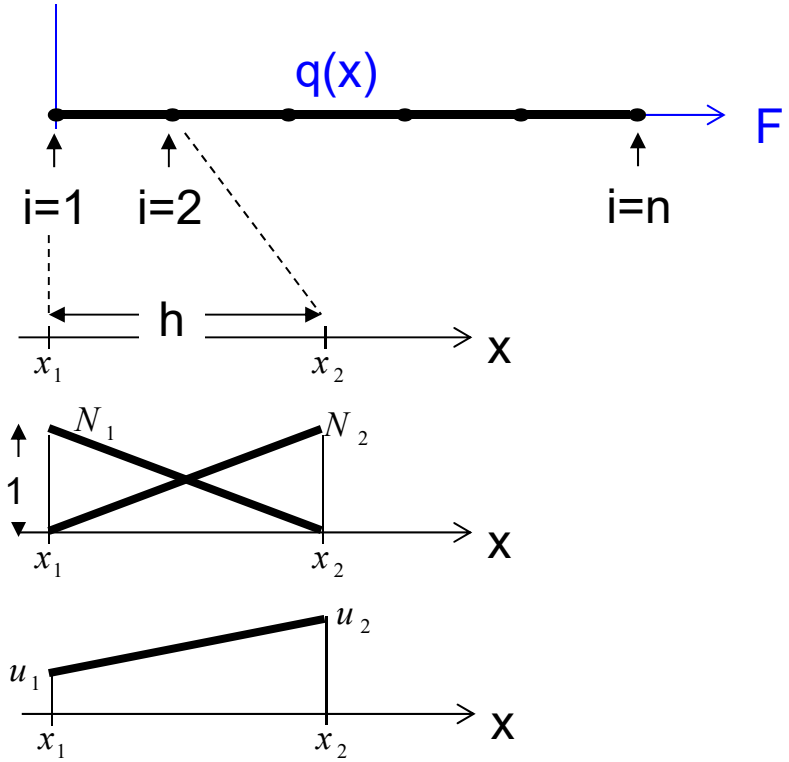
1. **Is the loading frequency well separated from the structure's natural frequency of vibrations?**
   (Example: design of bridges.)

2. **What is the time-history response?**
   (Example: car crash simulations.)





Example, finite elements

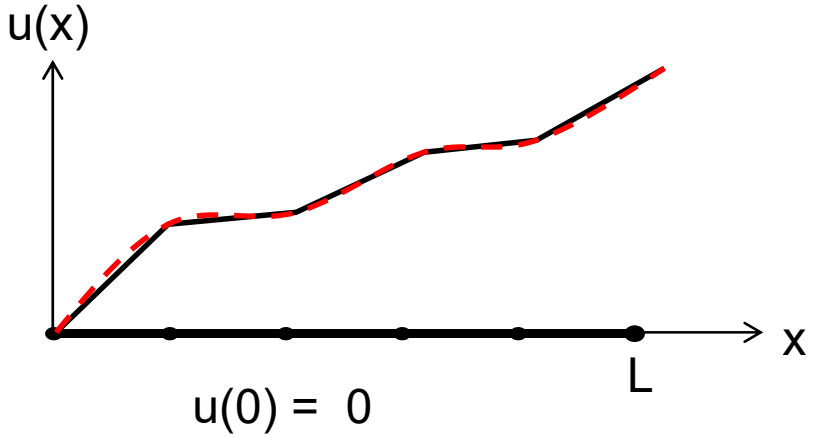


Assume piecewise linear:

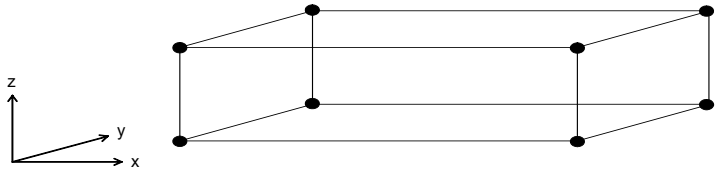
$$u(x) = \sum_{i=1}^{n} u_i N_i(x)$$

u(x) at the nodes

Element shape function



u(0) = 0

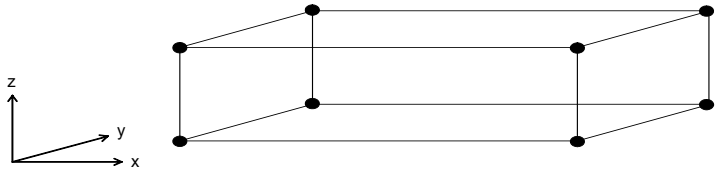


Displacements

$$u^{(m)}(x, y, z) = N^{(m)}(x, y, z)U$$

$$U = \begin{bmatrix} u_{1x} \\ u_{1y} \\ \\ \\ u_{n\ xrot} \\ \\ \\ \end{bmatrix}$$

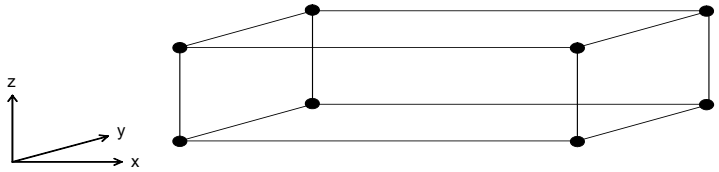




Velocities

$$\dot{u}^{(m)}(x, y, z) = N^{(m)}(x, y, z)\dot{U}$$

$$\dot{U} = \begin{bmatrix} \dot{u}_{1x} \\ \dot{u}_{1y} \\ \\ \dot{u}_{n\ xrot} \\ \\ \end{bmatrix}$$

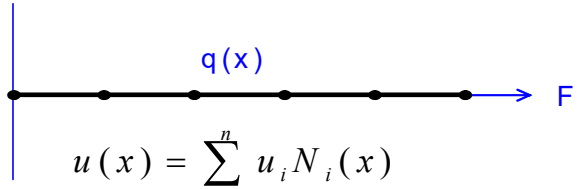


Accelerations

$$\ddot{u}^{(m)}(x, y, z) = N^{(m)}(x, y, z)\ddot{U}$$

$$\ddot{U} = \begin{bmatrix} \ddot{u}_{1x} \\ \ddot{u}_{1y} \\ \\ \ddot{u}_{n\ xrot} \\ \\ \end{bmatrix}$$

The load vector



$$u(x) = \sum_{i=1}^{n} u_i N_i(x)$$

$$\mathbf{KU} = \mathbf{F}$$

External forces

$$\sum_{i,j=1}^{n} \int_0^L EA \frac{dN_i}{dx}\frac{dN_j}{dx}dx \cdot u_i = \int_0^L qN_j dx + F(L)N_j(L) - F(0)N_j(0)$$

for every *i* and *j*

Static->dynamic: adding inertia and damping

$$\mathbf{KU} = \mathbf{F}$$

The load vector **F**: $\mathbf{F} = \mathbf{F}_B + \mathbf{F}_S + \mathbf{F}_C$

Element body forces:

$$\boldsymbol{F}_B = \sum_m \int_{V^{(m)}} \boldsymbol{N}^{(m)T} \boldsymbol{f}^{B(m)} \boldsymbol{dV}^{(m)}$$

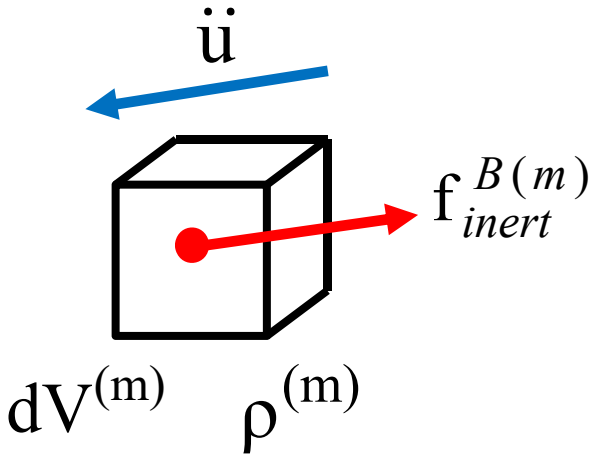


Static->dynamic: adding inertia and damping

Inertia forces are proportional to the acceleration (Newtons 2nd), and works in opposite direction:

$$\mathbf{f}_{inert}^{B(m)} = -\mathrm{m}^{(m)} \cdot \ddot{u}$$

$$= -\rho^{(m)} \cdot \mathrm{dV}^{(m)} \cdot \mathrm{N}^{(m)} \ddot{\mathrm{U}}$$



Including inertia to body forces:

$$\mathrm{F_B} = \sum_{m} \int_{V^{(m)}} \mathrm{N}^{(m)T} [\mathbf{f}^{B(m)} - \rho^{(m)} \mathrm{N}^{(m)} \ddot{\mathrm{U}}] \mathrm{dV}^{(m)}$$

Static->dynamic: adding inertia and damping

$$\mathbf{KU} = \mathbf{F}$$

The load vector **F**: $\mathbf{F} = \mathbf{F}_B + \mathbf{F}_S - \mathbf{F}_I + \mathbf{F}_C$

Including inertia to body forces:

$$F_B = \sum_m \int_{V^{(m)}} N^{(m)T} [f^{B(m)} - \rho^{(m)} N^{(m)} \ddot{U}] dV^{(m)}$$

$$F_B = \sum_m \int_{V^{(m)}} N^{(m)T} f^{B(m)} dV^{(m)} - \sum_m \int_{V^{(m)}} N^{(m)T} \rho^{(m)} N^{(m)} \ddot{U} dV^{(m)}$$



Static->dynamic: adding inertia and damping

Including inertia to body forces:

$$F_B = \sum_m \int_{V^{(m)}} N^{(m)T} f^{B(m)} dV^{(m)} - \sum_m \int_{V^{(m)}} N^{(m)T} \rho^{(m)} N^{(m)} \ddot{U} dV^{(m)}$$

$$KU = F - \underbrace{\sum_m \int_{V^{(m)}} N^{(m)T} \rho^{(m)} N^{(m)} dV^{(m)}}_{M} \ddot{U}$$

$$\mathbf{KU} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$



Dynamic analyses

$$\mathbf{K}\mathbf{U} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$



$$\boldsymbol{M} = \sum_{\boldsymbol{m}} \int_{\boldsymbol{V}^{(m)}} \rho^{(m)} \boldsymbol{N}^{(m)T} \boldsymbol{N}^{(m)} \boldsymbol{dV}^{(m)}$$

Dynamic analyses

$$\mathbf{K}\mathbf{U} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$



Static->dynamic: adding inertia and damping

$$\mathbf{KU} = \mathbf{F}$$

The load vector **F**: $\mathbf{F} = \mathbf{F}_B + \mathbf{F}_S - \mathbf{F}_I + \mathbf{F}_C$

Including damping to body forces:

$$F_B = \sum_m \int_{V^{(m)}} N^{(m)T} \left[f^{B(m)} - \rho^{(m)} N^{(m)} \ddot{U} - \kappa^{(m)} N^{(m)} \dot{U} \right] dV^{(m)}$$



Dynamic analyses

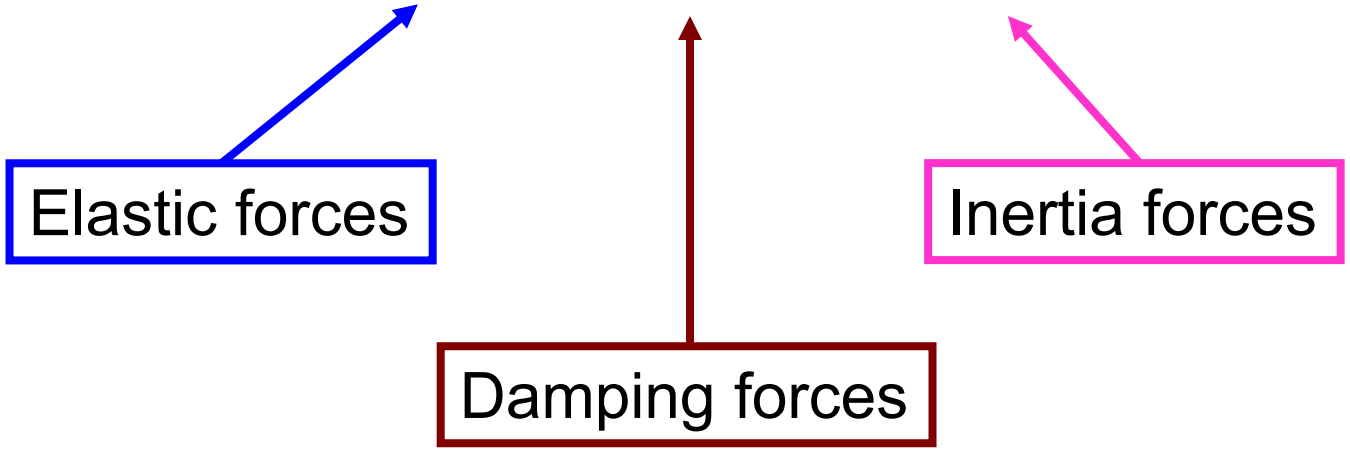
$$\mathbf{K}\mathbf{U} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$



$$\mathrm{C} = \sum_{\mathrm{m}} \int_{\mathrm{V}^{(\mathrm{m})}} \kappa^{(\mathrm{m})}\, \mathrm{N}^{(\mathrm{m})\mathrm{T}}\, \mathrm{N}^{(\mathrm{m})}\, \mathrm{dV}^{(\mathrm{m})}$$

Dynamic analyses

$$\mathbf{K}\mathbf{U} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$

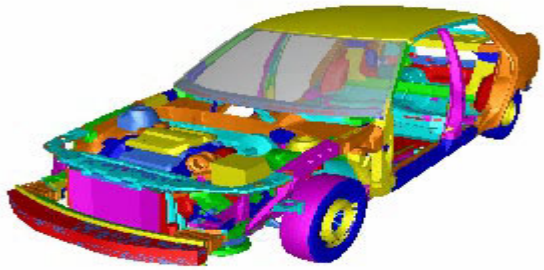


When is a problem dynamic?

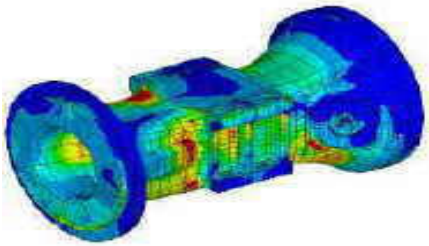
- If mass inertia effects the responses.
- Frequency of applied loading > $\omega/5$

1. **Is the loading frequency well separated from the structure's natural frequency of vibrations?**
   (Example: design of bridges.)

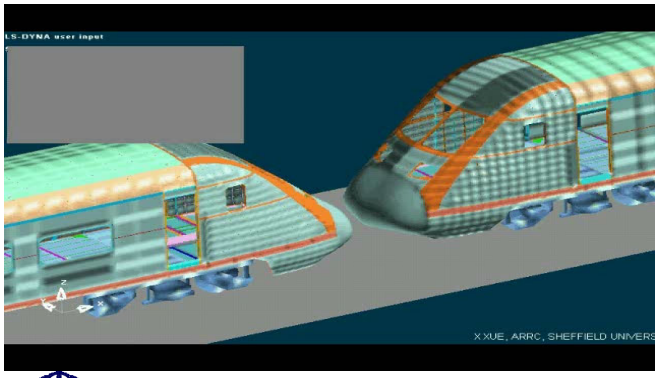
2. **What is the time-history response?**
   (Example: car crash simulations.)



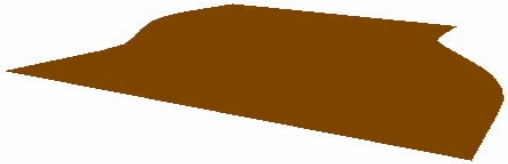




	DYNAMIC	STATIC
Structural analysis		
Crack propagation		
Accident reconstruction		
Drop tests (cell phones)		
Sheet metal forming		







Finite Element Analysis

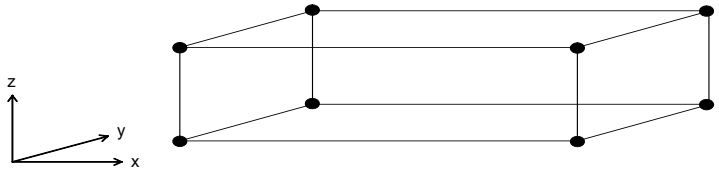
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- **Implicit Time Integration**



Dynamic Finite Element Analysis

- Explicit Time Integration
- Implicit Time Integration





Direct integration

$$\mathbf{K}\mathbf{U} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$

Integration is performed using a step-by-step method.

Satisfy equation at discrete time intervals, Δt apart.

Time derivatives are approximated with finite differences of displacements at different times.

Explicit

$$U_{t+\Delta t} = f(U_t, \dot{U}_t, \ddot{U}_t, U_{t-\Delta t}, \ldots)$$

Implicit

$$U_{t+\Delta t} = f(\dot{U}_{t+\Delta t}, \ddot{U}_{t+\Delta t}, U_t, \ldots)$$

Explicit / Implicit time integration

$$\mathbf{KU} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$

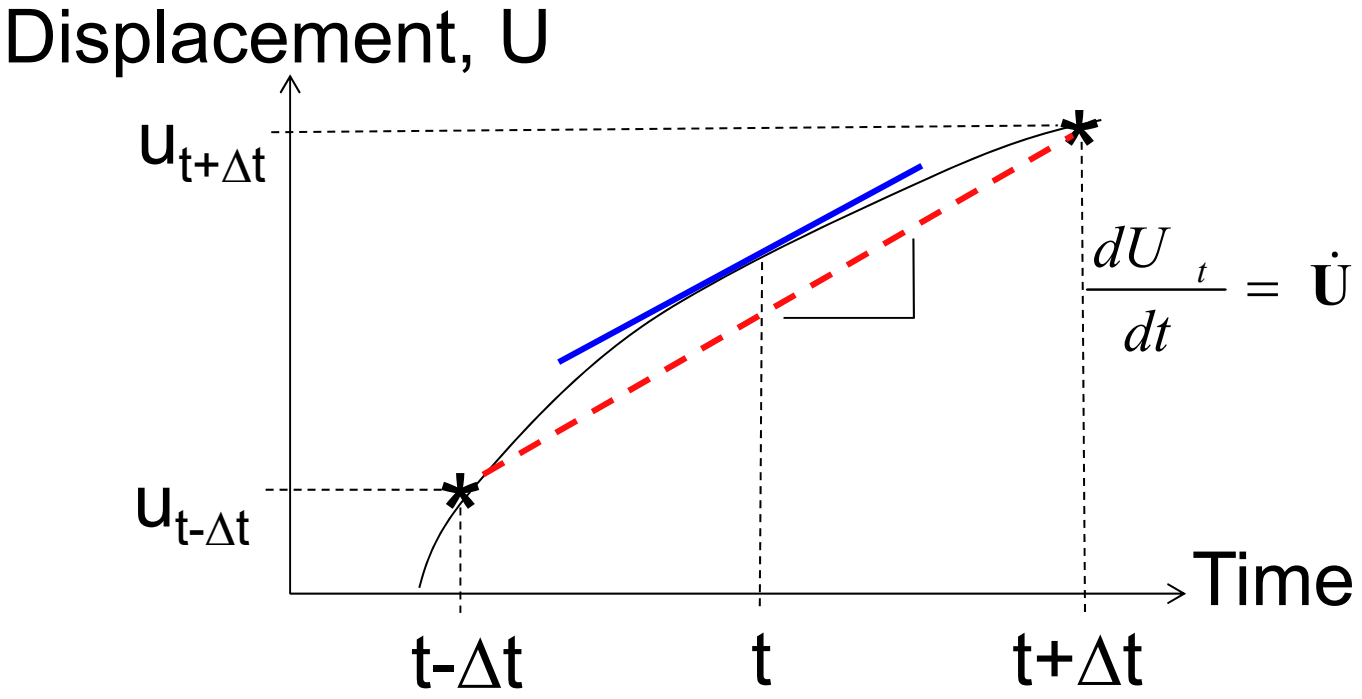
Explicit	Implicit
$U_{t+\Delta t} = f(U_t, \dot{U}_t, \ddot{U}_t, U_{t-\Delta t}, \ldots)$	$U_{t+\Delta t} = f(\dot{U}_{t+\Delta t}, \ddot{U}_{t+\Delta t}, U_t, \ldots)$
Equilibrium at time ***t***	Equilibrium at time ***t+Δt***
$\mathbf{KU}_t + \mathbf{C}\dot{\mathbf{U}}_t + \mathbf{M}\ddot{\mathbf{U}}_t = \mathbf{F}_t$	$\mathbf{KU}_{t+\Delta t} + \mathbf{C}\dot{\mathbf{U}}_{t+\Delta t} + \mathbf{M}\ddot{\mathbf{U}}_{t+\Delta t} = \mathbf{F}_{t+\Delta t}$
Conditionally stable: $\Delta \mathbf{t}_{\mathbf{crit}}$	Unconditionally stable
Central difference method	*Newmark's method*



Explicit Time Integration



Explicit – Central difference method



$$\frac{dU_t}{dt} = \dot{\mathbf{U}}_t = \frac{U_{t+\Delta t} - U_{t-\Delta t}}{2\Delta t}$$

$$\dot{\mathbf{U}}_t = \frac{1}{2\Delta t}\left(\mathbf{U}_{t+\Delta t} - \mathbf{U}_{t-\Delta t}\right) \qquad \ddot{\mathbf{U}}_t = \frac{1}{\Delta t^2}\left(\mathbf{U}_{t+\Delta t} - 2\mathbf{U}_t + \mathbf{U}_{t-\Delta t}\right)$$

Taylor expansion

Expanding $U_{t+\Delta t}$ and $U_{t-\Delta t}$ in Taylor series about time *t*

$$U_{t+\Delta t} = U_t + \Delta t\, \dot{U}_t + \frac{\Delta t^2}{2}\ddot{U}_t + \frac{\Delta t^3}{6}\dddot{U}_t + \cdots \qquad [1]$$

$$U_{t-\Delta t} = U_t - \Delta t\, \dot{U}_t + \frac{\Delta t^2}{2}\ddot{U}_t - \frac{\Delta t^3}{6}\dddot{U}_t + \cdots \qquad [2]$$

[1-2]
$$\dot{U}_t = \frac{1}{2\Delta t}\left(U_{t+\Delta t} - U_{t-\Delta t}\right) + O(\Delta t^2)$$

[1+2]
$$\ddot{U}_t = \frac{1}{\Delta t^2}\left(U_{t+\Delta t} - 2U_t + U_{t-\Delta t}\right) + O(\Delta t^2)$$



Explicit – Central difference method

$$\dot{\mathbf{U}}_t = \frac{1}{2\Delta t}\left(-\mathbf{U}_{t-\Delta t} + \mathbf{U}_{t+\Delta t}\right)$$

$$\ddot{\mathbf{U}}_t = \frac{1}{\Delta t^2}\left(\mathbf{U}_{t-\Delta t} - 2\mathbf{U}_t + \mathbf{U}_{t+\Delta t}\right)$$

$$\mathbf{K}\mathbf{U}_t + \mathbf{C}\dot{\mathbf{U}}_t + \mathbf{M}\ddot{\mathbf{U}}_t = \mathbf{F}_t$$

$$\mathbf{K}\mathbf{U}_t + \mathbf{C}\left(\frac{\mathbf{U}_{t+\Delta t} - \mathbf{U}_{t-\Delta t}}{2\Delta t}\right) + \mathbf{M}\left(\frac{\mathbf{U}_{t+\Delta t} - 2\mathbf{U}_t + \mathbf{U}_{t-\Delta t}}{\Delta t^2}\right) = \mathbf{F}_t$$

$$(a_0\mathbf{M} + a_1\mathbf{C})\mathbf{U}_{t+\Delta t} = \mathbf{F}_t - (\mathbf{K} - a_2\mathbf{M})\mathbf{U}_t - (a_0\mathbf{M} - a_1\mathbf{C})\mathbf{U}_{t-\Delta t}$$

Explicit – Central difference method

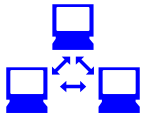
$$(a_0\mathbf{M} + a_1\mathbf{C})\mathbf{U}_{t+\Delta t} = \mathbf{F}_t - (\mathbf{K} - a_2\mathbf{M})\mathbf{U}_t - (a_0\mathbf{M} - a_1\mathbf{C})\mathbf{U}_{t-\Delta t}$$

If diagonal:

Known values

$$\begin{bmatrix} X & 0 & 0 & \cdots & 0 \\ 0 & X & 0 & \cdots & 0 \\ 0 & 0 & X & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & 0 & X \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} X \\ X \\ X \\ \vdots \\ X \end{bmatrix}$$

No equations need to be solved!



If not diagonal the method is no longer explicit.



The critical time step

Compute Δt^{e}_{crit}

$$\Delta t_{crit} \leq \min \frac{l_e}{c_e}$$

Characteristic length of element

Current wave speed in element

$$c_e = \sqrt{E / \rho}$$

$\Delta t = \alpha \Delta t_{crit}$ $\qquad 0.67 \leq \alpha \leq 0.98$



Implicit Time Integration



Implicit – Newmark's method

1. $\mathbf{K}\mathbf{U}_{t+\Delta t} + \mathbf{C}\dot{\mathbf{U}}_{t+\Delta t} + \mathbf{M}\ddot{\mathbf{U}}_{t+\Delta t} = \mathbf{F}_{t+\Delta t}$

2. $\dot{\mathbf{U}}_{t+\Delta t} = \dot{\mathbf{U}}_t + \Delta t\left((1-\gamma)\ddot{\mathbf{U}}_t + \gamma\ddot{\mathbf{U}}_{t+\Delta t}\right)$

3. $\mathbf{U}_{t+\Delta t} = \mathbf{U}_t + \Delta t\,\dot{\mathbf{U}}_t + \frac{\Delta t^2}{2}\left((1-2\beta)\ddot{\mathbf{U}}_t + 2\beta\,\ddot{\mathbf{U}}_{t+\Delta t}\right)$

SOLUTION:

Eq. 3 $\Rightarrow$ $\ddot{\mathbf{U}}_{t+\Delta t} = f\left(\mathbf{U}_t, \mathbf{U}_{t+\Delta t}, \dot{\mathbf{U}}_t, \ddot{\mathbf{U}}_t\right)$

↓

in Eq. 2 $\Rightarrow$ $\dot{\mathbf{U}}_{t+\Delta t} = f\left(\mathbf{U}_t, \mathbf{U}_{t+\Delta t}, \dot{\mathbf{U}}_t, \ddot{\mathbf{U}}_t\right)$

↓

Solve Eq. 1 for $\mathbf{U}_{t+\Delta t}$



Implicit – Newmark's method

2. $$\dot{\mathbf{U}}_{t+\Delta t} = \dot{\mathbf{U}}_t + \Delta t\left((1-\gamma)\ddot{\mathbf{U}}_t + \gamma\ddot{\mathbf{U}}_{t+\Delta t}\right)$$

3. $$\mathbf{U}_{t+\Delta t} = \mathbf{U}_t + \Delta t\,\dot{\mathbf{U}}_t + \frac{\Delta t^2}{2}\left((1-2\beta)\ddot{\mathbf{U}}_t + 2\beta\,\ddot{\mathbf{U}}_{t+\Delta t}\right)$$

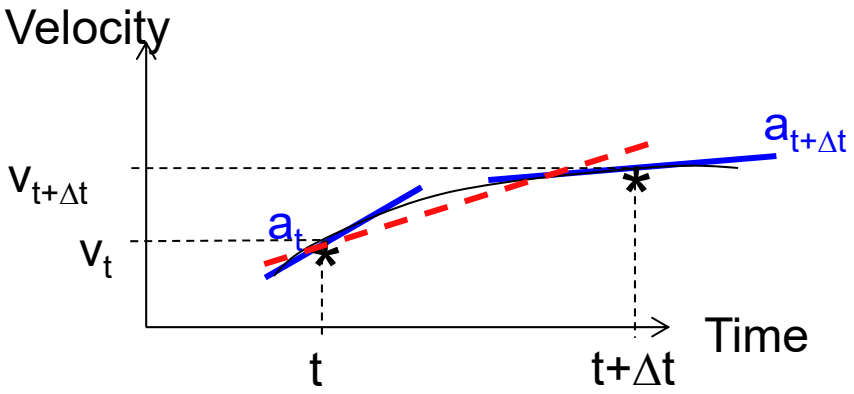
$\left.\begin{array}{l}\gamma = \frac{1}{2}\\ \beta = \frac{1}{4}\end{array}\right\} \Rightarrow$ Unconditionally stable

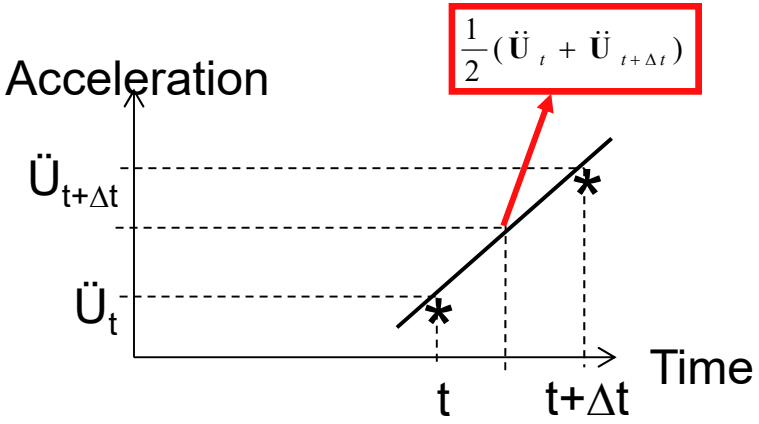
$\left.\begin{array}{l}\gamma = \frac{1}{2}\\ \beta = 0\end{array}\right\} \Rightarrow$ Central difference method



Implicit – Newmark's method

2. $\dot{\mathbf{U}}_{t+\Delta t} = \dot{\mathbf{U}}_t + \Delta t\left((1-\gamma)\ddot{\mathbf{U}}_t + \gamma\ddot{\mathbf{U}}_{t+\Delta t}\right)$



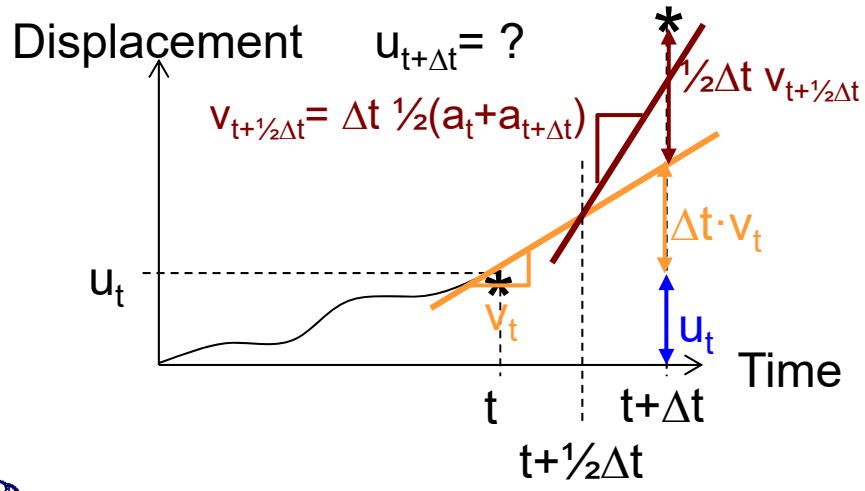


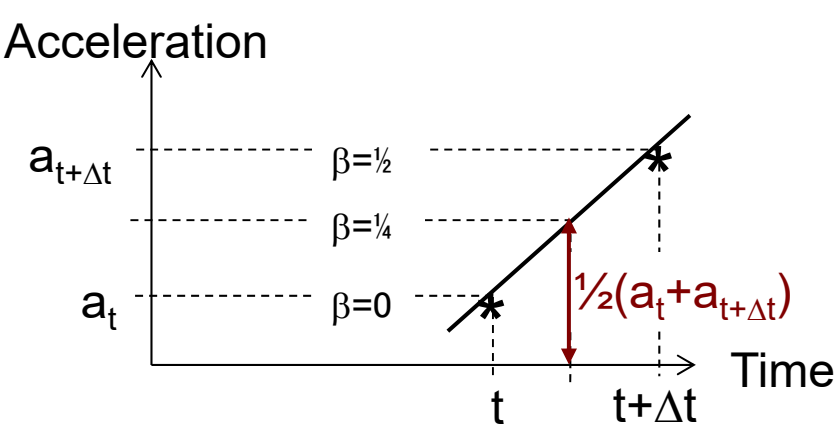
$$\gamma = \frac{1}{2} \quad \Rightarrow \quad \dot{\mathbf{U}}_{t+\Delta t} = \dot{\mathbf{U}}_t + \Delta t\left(\frac{1}{2}\ddot{\mathbf{U}}_t + \frac{1}{2}\ddot{\mathbf{U}}_{t+\Delta t}\right)$$

Implicit – Newmark's method

3. $$\mathbf{U}_{t+\Delta t} = \mathbf{U}_t + \Delta t\,\dot{\mathbf{U}}_t + \frac{\Delta t^2}{2}\left((1-2\beta)\ddot{\mathbf{U}}_t + 2\beta\,\ddot{\mathbf{U}}_{t+\Delta t}\right)$$

$$\beta = \frac{1}{4} \Rightarrow \mathbf{U}_{t+\Delta t} = \mathbf{U}_t + \Delta t\,\dot{\mathbf{U}}_t + \frac{\Delta t}{2}\Delta t\left(\frac{1}{2}\dot{\mathbf{U}}_t + \frac{1}{2}\dot{\mathbf{U}}_{t+\Delta t}\right)$$





Implicit – Newmark's method

1. $$\mathbf{K}\mathbf{U}_{t+\Delta t} + \mathbf{C}\dot{\mathbf{U}}_{t+\Delta t} + \mathbf{M}\ddot{\mathbf{U}}_{t+\Delta t} = \mathbf{F}_{t+\Delta t}$$

2. $$\dot{\mathbf{U}}_{t+\Delta t} = \dot{\mathbf{U}}_t + \Delta t\left((1-\gamma)\ddot{\mathbf{U}}_t + \gamma\ddot{\mathbf{U}}_{t+\Delta t}\right)$$

3. $$\mathbf{U}_{t+\Delta t} = \mathbf{U}_t + \Delta t\,\dot{\mathbf{U}}_t + \frac{\Delta t^2}{2}\left((1-2\beta)\ddot{\mathbf{U}}_t + 2\beta\,\ddot{\mathbf{U}}_{t+\Delta t}\right)$$

SOLUTION:

Eq. 3 $\Rightarrow$ $\ddot{\mathbf{U}}_{t+\Delta t} = f\left(\mathbf{U}_t, \mathbf{U}_{t+\Delta t}, \dot{\mathbf{U}}_t, \ddot{\mathbf{U}}_t\right)$

↓

in Eq. 2 $\Rightarrow$ $\dot{\mathbf{U}}_{t+\Delta t} = f\left(\mathbf{U}_t, \mathbf{U}_{t+\Delta t}, \dot{\mathbf{U}}_t, \ddot{\mathbf{U}}_t\right)$

↓

Solve Eq. 1 for $\mathbf{U}_{t+\Delta t}$



Explicit / Implicit time integration

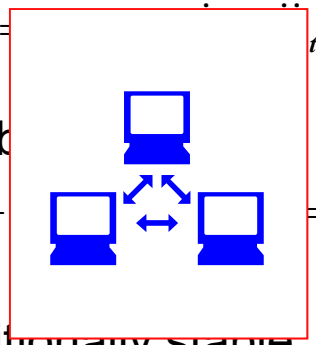
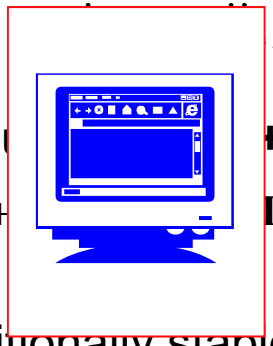
$$\mathbf{KU} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$

Explicit	Implicit
$U_{t+\Delta t} = f(U_t, \dot{U}_t, \ddot{U}_t, U_{t-\Delta t}, \ldots)$	$U_{t+\Delta t} = f(\dot{U}_{t+\Delta t}, \ddot{U}_{t+\Delta t}, U_t, \ldots)$
Equilibrium at time t	Equilibrium at time $t+\Delta t$
$\mathbf{KU}_t + \mathbf{C}\dot{\mathbf{U}}_t + \mathbf{M}\ddot{\mathbf{U}}_t = \mathbf{F}_t$	$\mathbf{KU}_{t+\Delta t} + \mathbf{C}\dot{\mathbf{U}}_{t+\Delta t} + \mathbf{M}\ddot{\mathbf{U}}_{t+\Delta t} = \mathbf{F}_{t+\Delta t}$
Conditionally stable: $\Delta \mathbf{t}_{\mathbf{crit}}$	Unconditionally stable
Central difference method	Newmark's method



Explicit / Implicit time integration

$$\mathbf{KU} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$

Explicit	Implicit
$U_{t+\Delta t} = \ldots(U_t, U_{t-\Delta t}, \ldots)$	$U_{t+\Delta t} = \ldots(U_{t+\Delta t}, U_t, \ldots)$
Equilib…	Equilibri… $+\Delta t$
$\mathbf{KU}_t + \ldots = \mathbf{F}_t$	$\mathbf{KU}_{t+\Delta t} + \ldots \mathbf{M}\ddot{\mathbf{U}}_{t+\Delta t} = \mathbf{F}_{t+\Delta t}$
	
Conditionally stable: Δt_{crit}	Unconditionally stable
Central difference method	Newmark's method



Explicit / Implicit time integration

$$\mathbf{KU} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{M}\ddot{\mathbf{U}} = \mathbf{F}$$

Explicit

+No equations need to be solved, very simple and trouble free algorithm, complex phenomena easily included.
+accuracy is assured if Δt stable for large systems
+no stiffness matrix necessary – saves storage
-conditionally stable, small Δt

Impact, crash (ms)
Wave propagation

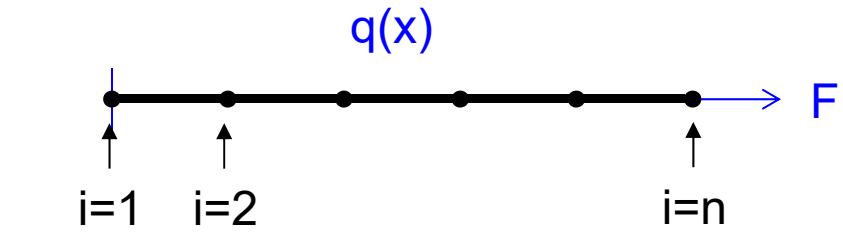
Implicit

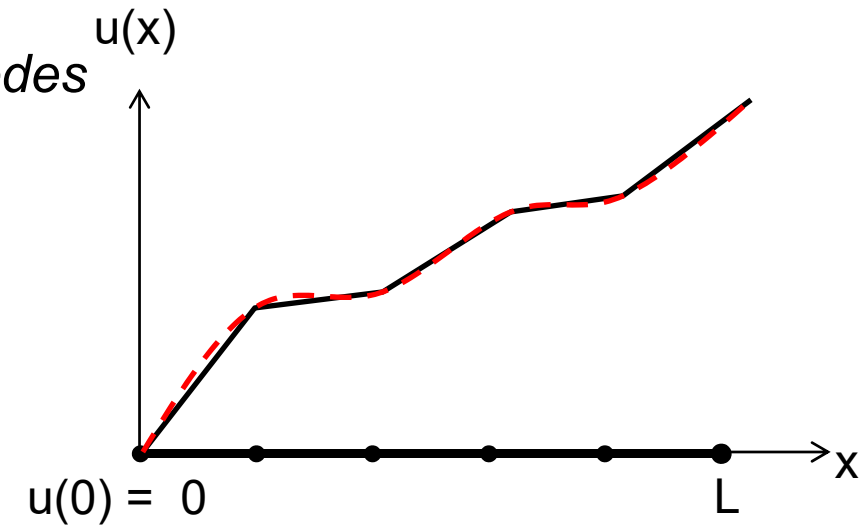
+unconditionally stable -> *large* Δt *possible*
-accuracy can deteriorate for large Δt
-equations must be solved, complex algorithm with lower reliability in non-linear situations
-Newton form has large storage requirements

Long duration impacts (s), quasistatic and large deformations.
RBD (rigid body dynamics) often implicit -> large Δt *possible*

Numerical errors *Discretization* - Convergence analysis

- Displacement error: $\propto h^2$
- Strain error: $\propto h$
- *Displacements most accurate at or near nodes*
- *Strains most accurate in element interior (at or near Gauss points)*





Numerical errors *Dynamics* - Energy balance

Check energy balance as a function of time

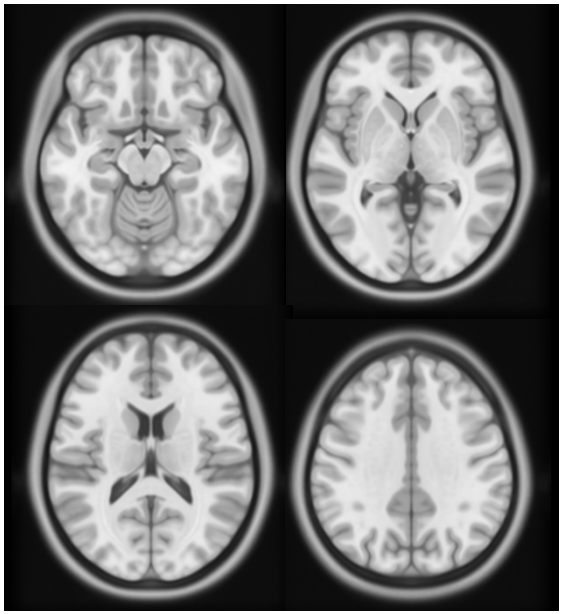
$$\left| W_{kin} + W_{\text{int}} - W_{ext} \right| \leq \varepsilon \max\left(W_{kin}, W_{\text{int}}, W_{ext} \right)$$

$\varepsilon \approx 10^{-2}$

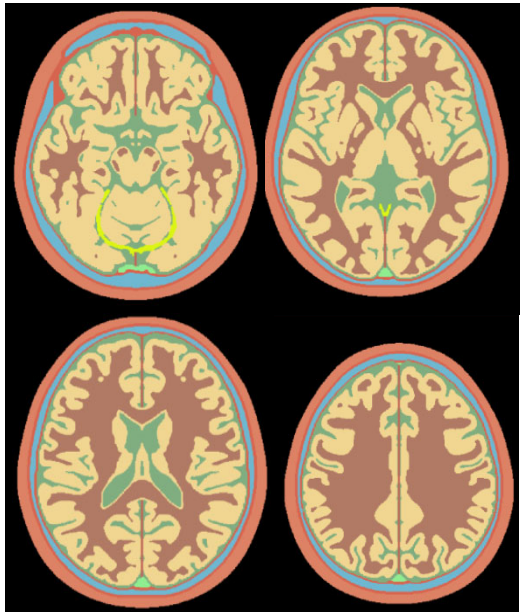
Biomechanics Applications of Dynamic Finite Element Analysis



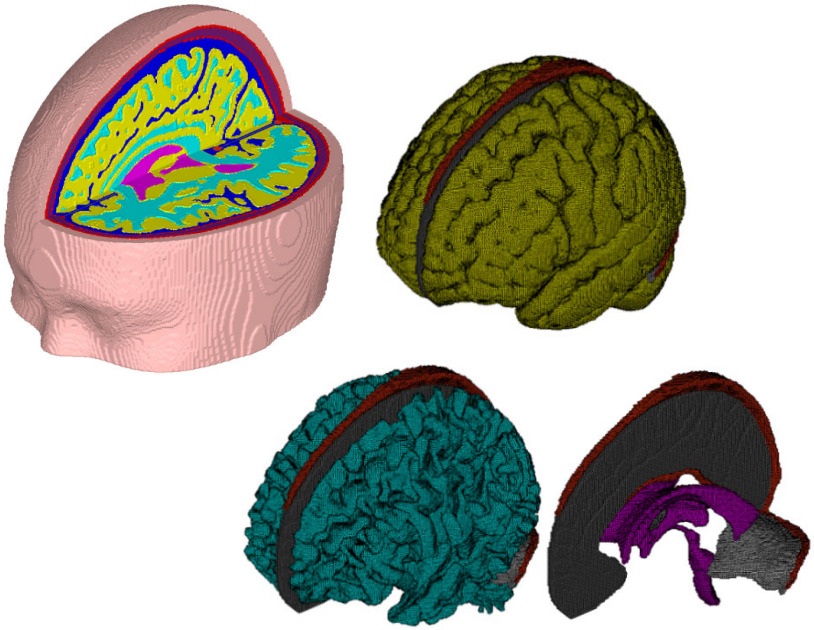
Subject specific TBI modelling



Input Images: MRI



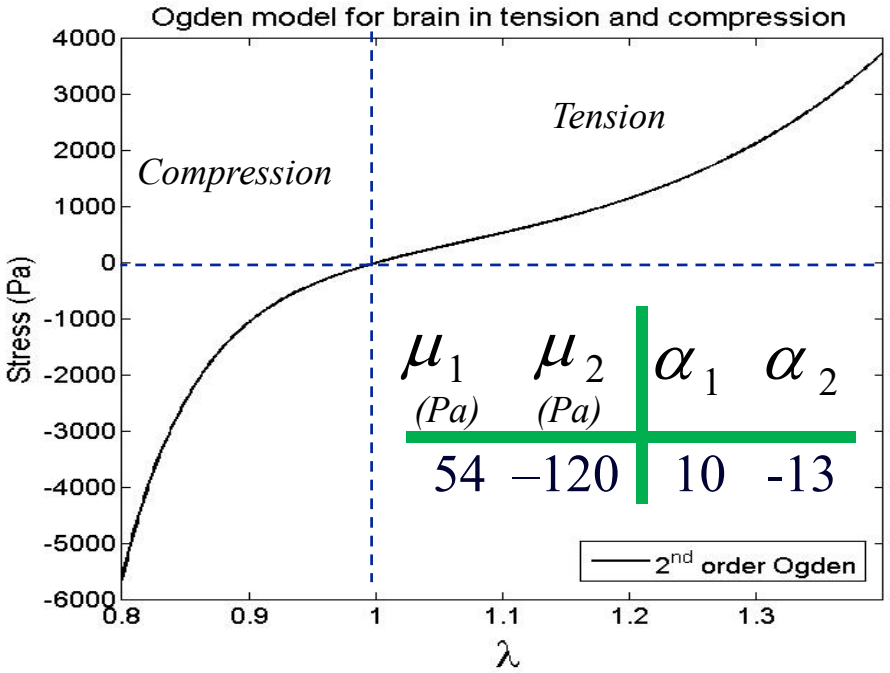
Segmented Res



Authentic subject specific FE-model

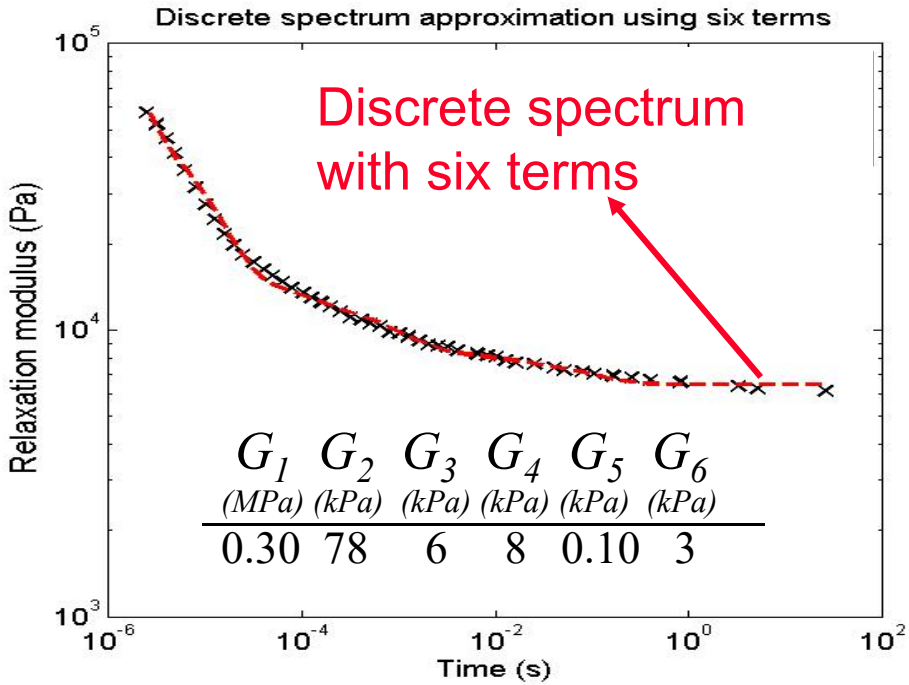
Ho *et al.* J. Biomech. 2017; Li *et al.* Biomechanics and modeling in mechanobiology, 2020

Non-linear viscoelastic brain properties



μ_1 (Pa)	μ_2 (Pa)	α_1	α_2
54	–120	10	-13

Data from: Franceschini, 2006



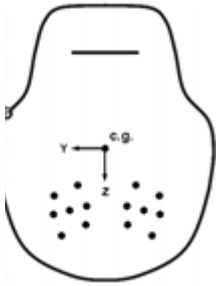
G_1 (MPa)	G_2 (kPa)	G_3 (kPa)	G_4 (kPa)	G_5 (kPa)	G_6 (kPa)
0.30	78	6	8	0.10	3

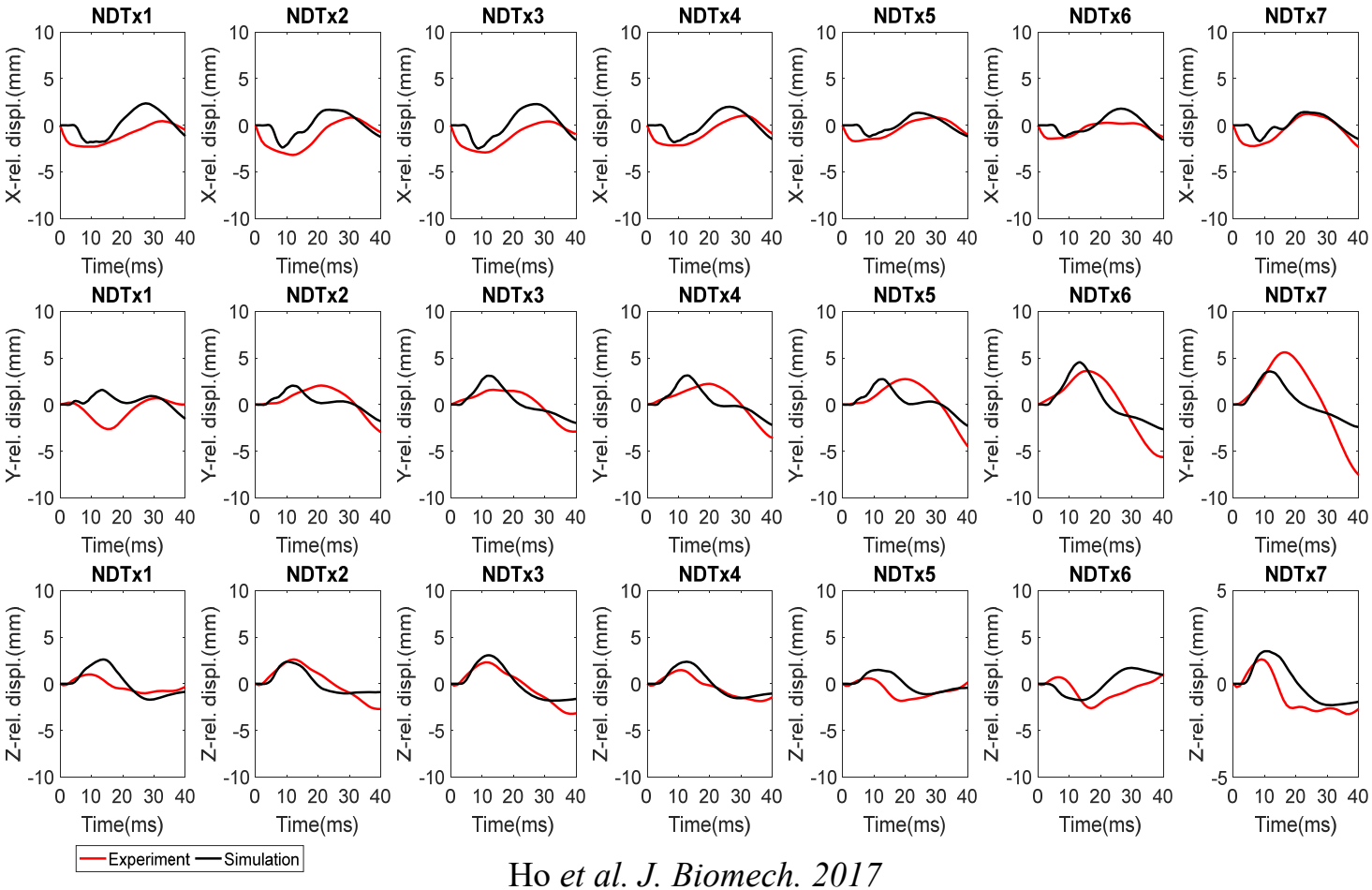
x *Nicolle et al., 2004*

Kleiven, 2007

β_1	β_2	β_3	β_4	β_5	β_6
10^6	10^5 …				10^1

Comparison with post-mortem brain motion (Hardy *et al.* 2007)

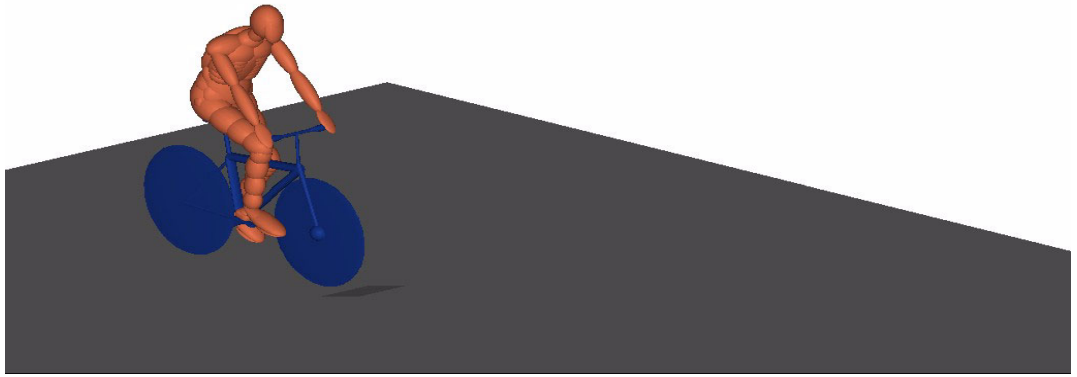




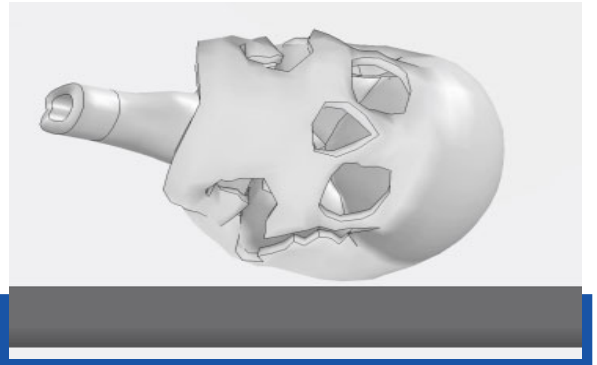
Ho *et al. J. Biomech. 2017*

Bicycle Accident Reconstruction

- Elderly man
- Uneven ground
- Skull fracture
  Contusion
- No Helmet

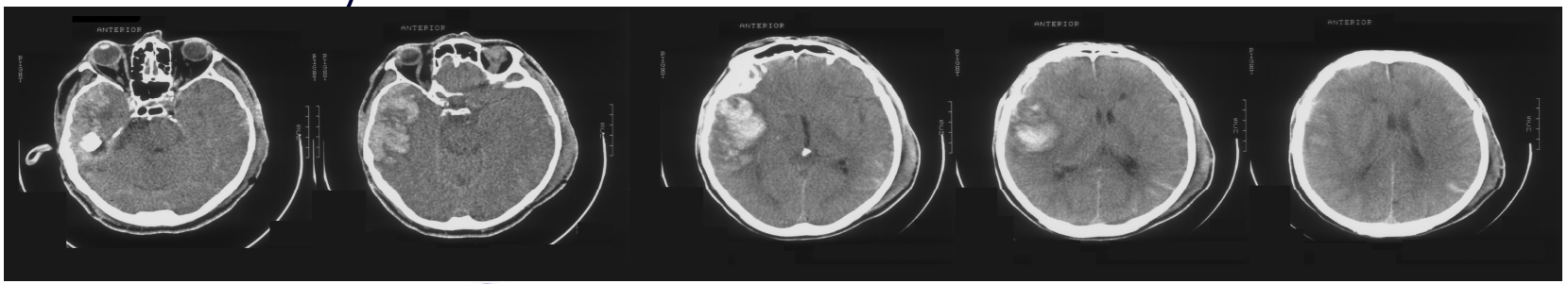






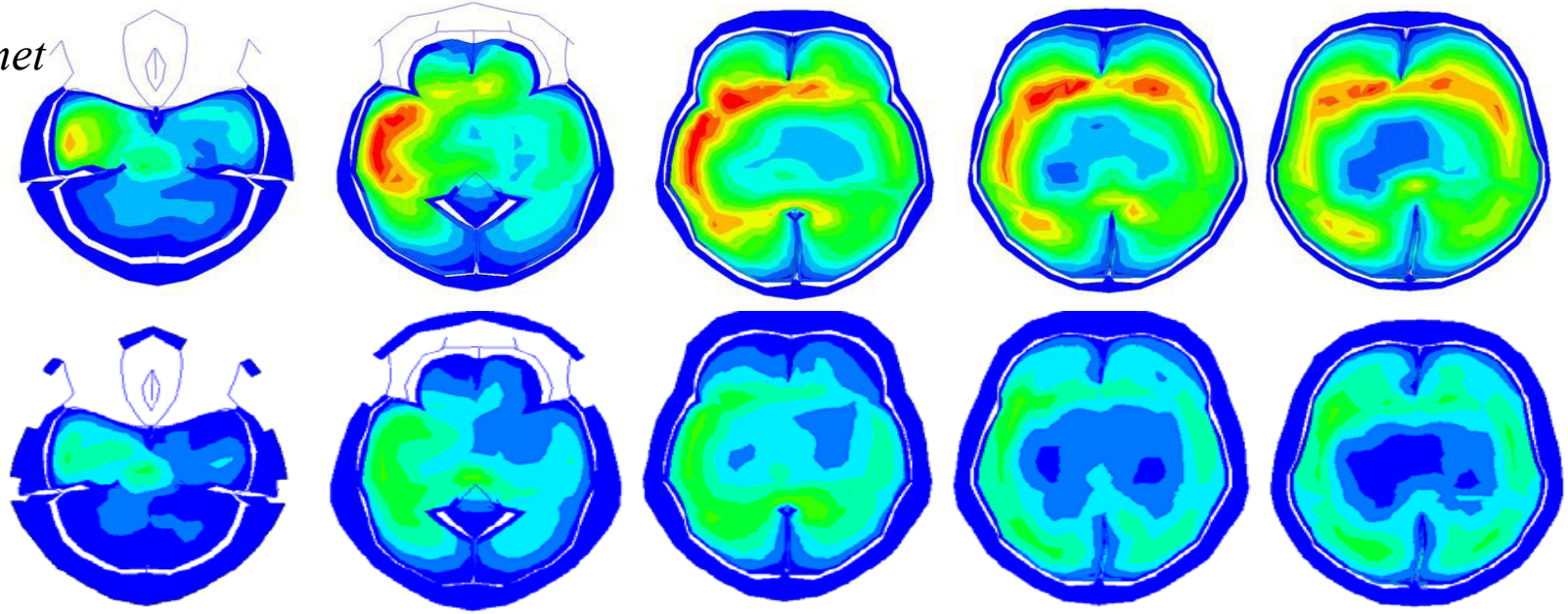
(Fahlstedt et al. J.Biomech. 2015)

Bicycle Accident reconstruction



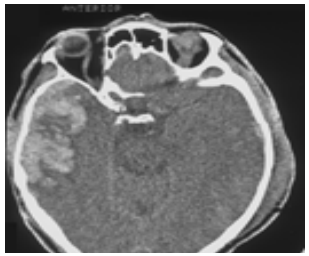
No helmet

Helmet

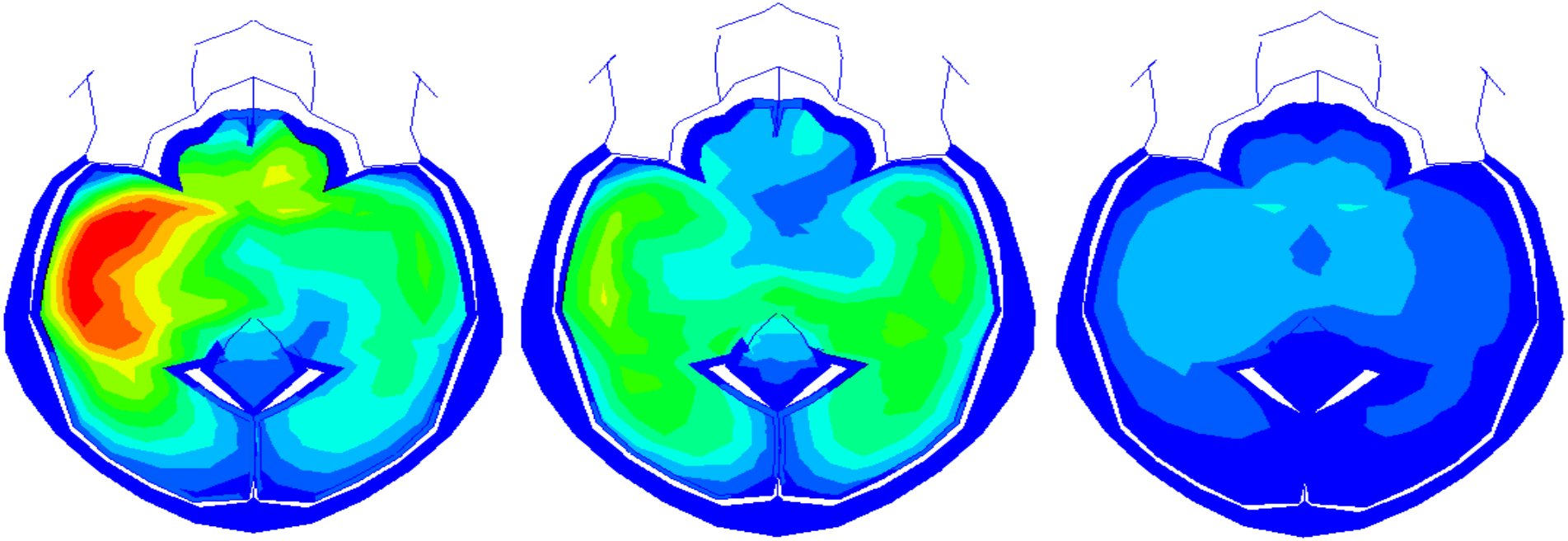


0 1st Principal Green-Lagrange Strain 0.3

Fahlstedt *et al.* (2016). Acc. Analys. & Prev



Bicycle Accident reconstruction



Impact *Only angular* *Only linear*

0 1st Principal Green-Lagrange Strain 0.3

Thank you!

sveink@kth.se